

Single Sample t -Test

ITAO 20200: Foundations of Statistics

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Answers to Practice On Your Own

```
# Importing the dataset and naming it "Grades".
Grades <- read.csv("https://mendoza-stats.s3.amazonaws.com/grades.csv")
head(Grades, n = 10) # First 10 observations in the dataset
```

##	Pretest	Midterm	Final	Teacher	Required
## 1	56	75	77	1	1
## 2	79	91	84	1	0
## 3	68	82	81	1	1
## 4	57	72	72	1	0
## 5	70	81	76	1	1
## 6	60	77	75	1	0
## 7	74	88	81	1	1
## 8	73	85	86	1	0
## 9	61	80	79	2	1
## 10	50	74	69	2	0

1. Midterm Scores

Obtain some summary statistics about **Midterm Scores** and determine if the average score is *less than* a population value of 84 using $\alpha = 0.05$, i.e

$$H_0 : \mu \geq 84$$

$$H_a : \mu < 84$$

Some summary statistics for Midterm Scores

```
summary(Grades$Midterm)
```

##	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
##	72.00	78.50	81.50	82.21	85.50	95.00

```
sd(Grades$Midterm)
```

```
## [1] 5.999849
```

```
length(Grades$Midterm)
```

```
## [1] 24
```

Single sample t -test for Midterm Scores

```
t.test(Grades$Midterm,          # the variable we are testing
       alternative = "less",    # the alternative hypothesis
       mu = 84,                 # the null (or test) value
       conf.level = 0.95)      # the confidence level = 1 - alpha
```

```
##
## One Sample t-test
##
## data: Grades$Midterm
## t = -1.4629, df = 23, p-value = 0.07851
## alternative hypothesis: true mean is less than 84
## 95 percent confidence interval:
##      -Inf 84.30734
## sample estimates:
## mean of x
## 82.20833
```

Decision & Conclusion

A single sample t test was performed to determine if average score on the Midterm was less than the expected value of 84. Results indicate that the average score on the Midterm ($\bar{x} = 82.21$, $s = 6$) was not significantly lower than expected, $t(23) = -1.5$, $p = 0.0785$.

2. Final Scores

Obtain some summary statistics about **Final Scores** and determine if the average score is *greater than* an expected value of 85 using $\alpha = 0.05$, i.e

$$H_0 : \mu \leq 85$$

$$H_a : \mu > 85$$

Some summary statistics for Final Scores

```
summary(Grades$Final)

##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   69.00   78.50   84.50   84.62   92.00  100.00

sd(Grades$Final)

## [1] 8.355225

length(Grades$Final)

## [1] 24
```

Single sample t -test for Final Scores

```
t.test(Grades$Final,           # the variable we are testing
       alternative = "greater", # the alternative hypothesis
       mu = 85,                # the null (or test) value
       conf.level = 0.95)      # the confidence level = 1 - alpha

##
## One Sample t-test
##
## data:  Grades$Final
## t = -0.21988, df = 23, p-value = 0.586
## alternative hypothesis: true mean is greater than 85
## 95 percent confidence interval:
##  81.70199      Inf
## sample estimates:
## mean of x
##    84.625
```

Decision & Conclusion

A single sample t -test was performed to determine if average score on the Final was greater than the expected value of 85. Results indicate that the average score on the Final ($\bar{x} = 84.62$, $s = 8.36$) was not significantly greater than expected, $t(23) = -0.22$, $p = 0.586$.